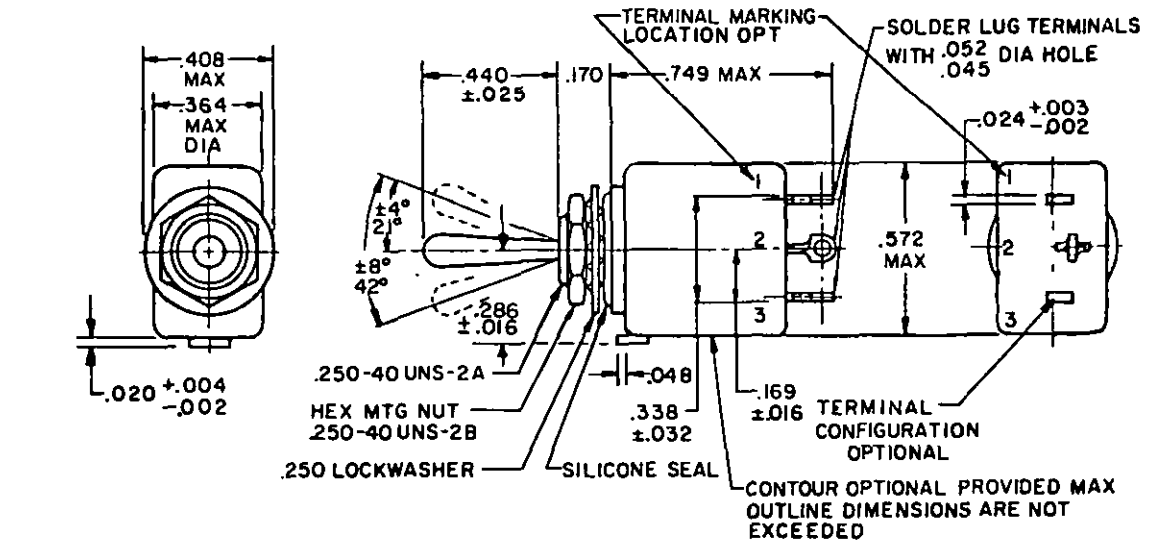
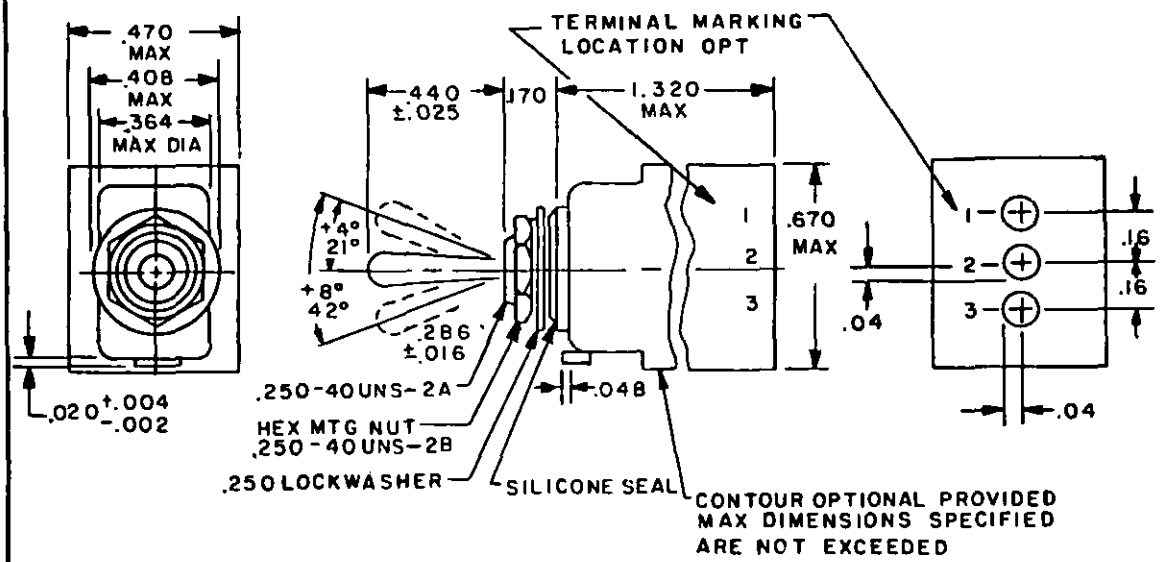
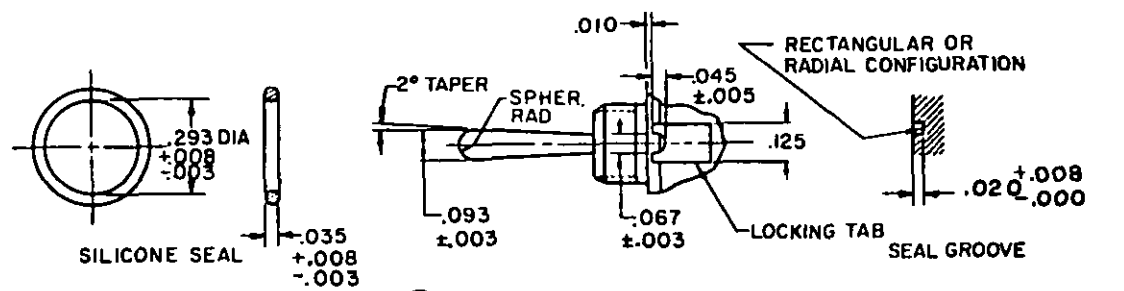
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(K) Denotes change

P.A AF-85 Other Cust ARMY-ER NAVY-AS	International Interest ASCCAS 12/16A	TITLE SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE TOGGLE SEALED, SINGLE POLE, SOLDER LUG, OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS	MILITARY STANDARD MS24655
Procurement Specification MIL-S-8834	SUPERSEDES: MS0024655 (USAF)		PAGE 1 OF 5

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User activities: Army - MI Navy - Air Force -

Review activities: Army - AR, AV Navy - EC Air Force - 11, 17, 99

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Review activities: Army - AR, AV
Navy - EC
Air Force - 11, 17, 99

User activities: Army - MI
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Procurement Specification	P.A	International Interest
MIL-S-8834	AF-85 Other Cust ARMY-ER NAVY-AS	ASCCAS 12/16A

SUPERSEDES: MS0024655 (USAF)

TITLE: SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE TOGGLE SEALED, SINGLE POLE, SOLDER LUG, OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS

MILITARY STANDARD

MS24655

DETAIL REQUIREMENTS: 5/

MS DASH NO	CIRCUITS MADE BETWEEN TERMINALS AS INDICATED WITH THE TOGGLE LEVER IN THESE POSITIONS: 1/			CURRENT CAPACITY IN AMPERES						LIFE LOW CURRENT LEVEL SWITCHING 5 mV
				RESISTIVE LOAD			INDUCTIVE LOAD			
					115 VOLTS			115 VOLTS		
	OPPOSITE LOCKING 4/ TAB SIDE	CENTER POSITION	LOCKING TAB SIDE 4/	28 VOLTS AC	60 HERTZ AC	400 HERTZ AC	28 VOLTS DC	60 HERTZ AC	400 HERTZ AC	
-211	2 - 3 ON	OFF	ON 1 - 2	5	2	3	2/	1	2	3/
-221		NONE	OFF							
-231		NONE	ON 1 - 2							
-241			NONE							
-271	MOM-ON 2-3	OFF	1 - 2 MOM-ON				1	1	2	25 µA
-281	NONE									
-311	ON 2 - 3									
-321 6/7/	NONE	ON								

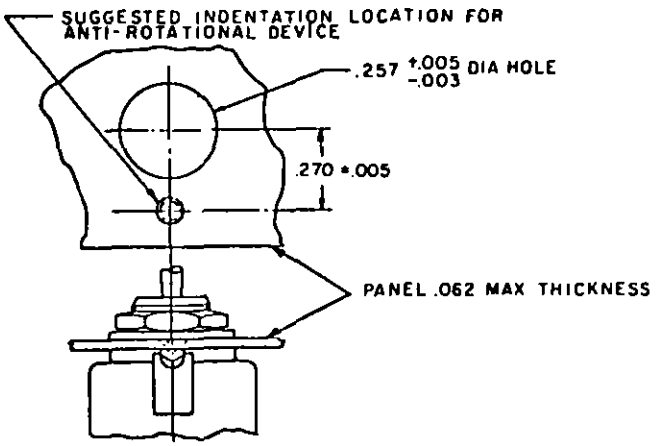
NOTES:
1/ Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
2/ With time constant of .020 ±.002 seconds.
3/ Contact resistance not to exceed 50 ohms during life, low current level switching.
4/ Nonfunctional terminals shall not be supplied.
5/ Part number example: For IWTS termination MS number will be concluded by the letter W. Example: MS24655-211 with IWTS termination will be MS24655-211W.
6/ Dielectric withstanding voltage: 1,200 V rms at sea level (center on circuits).
7/ Delayed action of the switch toggle lever may cause circuit to close or open before snap action mechanism trips.

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INCHES	MM	INCHES	MM	INCHES	MM
.002	0.05	.035	0.89	.257	6.53
.003	0.08	.040	1.02	.270	6.86
.004	0.10	.045	1.14	.286	7.26
.005	0.13	.048	1.22	.293	7.44
.008	0.20	.050	1.27	.338	8.59
.010	0.25	.062	1.57	.364	9.25
.016	0.41	.067	1.70	.408	10.36
.020	0.51	.093	2.36	.440	11.18
.024	0.61	.125	3.18	.470	11.94
.025	0.64	.160	4.06	.572	14.53
.028	0.71	.169	4.29	.670	17.02
.032	0.81	.170	4.32	.749	19.02
		.250	6.35	1.320	33.53

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User activities: Army - MI
Navy -
Air Force -

Review activities: Army - AR,AV
Navy - EC
Air Force - 11,17,99

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P.A	International Interest	TITLE	MILITARY STANDARD
AF - 85 Other Cust ARMY-ER NAVY-AS	ASCCAS 12/16A	SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE TOGGLE SEALED, SINGLE POLE, SOLDER LUG, OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS	MS 24655
Procurement Specification MIL-S-8834		SUPERSEDES: MS0024655 (USAF)	PAGE 3 OF 5

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(K) NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) on decimals and ±.5 on angles.
4. For hardware detail specifications see appendix of MIL-S-8834.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
7. Altitude 50,000 feet.
8. 115 V ac, 60 hertz electrical endurance tests are to be performed at room temperature and pressure.
9. Weight: .011 pound maximum (5.0 grams). For switches with solder lug terminals.
   .0211 pound maximum (9.6 grams). For switches with IWTS termination.
10. Shock: Switches with solder terminals, Method I and Method II (high impact). Switches with IWTS terminals, Method I only. The switch shall be electrically and mechanically operative at the conclusion of the test and there shall be no mechanical transfer during the test.
11. Suitable for mounting on panels of .062 (1.57 mm) maximum thickness.
12. Strength of terminal: 5 pound normal to the mounting planes, and 2 pounds in other planes, (solder lug terminals only).
13. Strength of actuator: Lever pivot and lever stop 6 pounds.
14. Switches with IWTS terminals shall accept M39029/1-101 connector contacts.
15. With IWTS termination, sealing plugs may be used in non-functional terminal grommet holes. Switches shall accept MS27488-20 end seal plugs.
16. The terminal sealing grommet for switches with IWTS terminals shall seal on smooth wire insulations of .040 (1.02 mm) to .083 (2.11 mm) diameter. The sealing grommet shall be color coded red to indicate contact size.
17. Installing/removal tool for IWTS connector contacts is M81969/14-02.

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P.A	International interest	TITLE	MILITARY STANDARD
AF-85 Other Cust ARMY-ER NAVY-AS	ASCCAS 12/16A	SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE TOGGLE SEALED, SINGLE POLE, SOLDER LUG, OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS	MS24655
Procurement Specification MIL-S-8834		SUPERSEDES: MS0024655 (USAF)	PAGE 4 OF 5

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The following test requirements apply only to those switches with integrated wire terminals:

TEST REQUIREMENTS:

Qualification and group B tests are to be performed in accordance with MIL-S-8834 except:

During all tests switches are to be fully wired with appropriate wire and terminal contacts.

Contact voltage drop: The contact voltage drop with two terminals and the switch contact in series shall not exceed 8 millivolts. This measurement shall be made from one wire contact through the switch contacts to the other wire contact with .1 ampere at a voltage of 2-4 volts DC.

Fluid immersion: Two additional qualification & group B switches, fully wired, shall be subjected to three exposure cycles in accordance with a and b.

a. The terminal end of the switch shall be immersed to a depth of 3/8-inch measured from the exposed face of the sealing grommet in each of the following fluids for 2 +1/2, -0 minutes between immersions. After each immersion the excess liquid is to be blown off the switch external surfaces with an air jet.

(1) MIL-T-5624 - Turbine fuel, aviation, grade JP-4 or JP-5.

(2) Skydrol 500A - Federal stock number 9150-857-9069.

(3) Coolanel - Federal stock number 9150-551-4022.

(4) MIL-E-9500 - Ethylene glycol, technical uninhibited.

(5) MIL-L-7808 - Lubricating oil, aircraft turbine engine synthetic base.

b. Exposure to ambient air for 24 ±2 hours.

c. At the end of the third cycle, the insulation resistance shall be measured and the switches shall be inspected for cracking and loosening of bonds and seams. When switches are tested as specified the insulation resistance shall not be less than 1,000 megohms and there shall be no evidence of cracking and loosening of bonds and seams.

User activities: Army - MI
Navy -
Air Force -

Review activities: Army - AR, AV
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P.A	International interest	TITLE	MILITARY STANDARD
AIR FORCE- 85 Other Cust ARMY-ER NAVY-AS	ASCCAS 12/16A	SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE TOGGLE SEALED, SINGLE POLE, SOLDER LUG, OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS	MS 24655
Procurement Specification MIL-S-8834		SUPERSEDES: MS0024655 (USAF)	PAGE 5 OF 5

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